

MeerK40t UI Manual (Meerkat)

MeerK40t — User Interface Manual (Meerkat workspace edition)

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MeerK40t — User Interface Manual

(Meerkat workspace edition)

Version: 1.1 (2026-06-03)

Based on: MeerK40t v**0.9.9000** on your PC, device **GRBL-DLC32-400**

Screenshots: images/meerk40t-ui-manual/ (copied from
Pictures\Screenshots\Meerk40T)

Audience: Andre van der Westhuizen — GRBL / MKS DLC32 CO2 workflow

About this manual

This document explains **what you see and click** in MeerK40t. It combines **source-code accuracy** with **your real screenshots** (June 2026 capture session).

Screenshot files

Location	Contents
images/meerk40t-ui-manual/*.png	28 named images used in this manual
images/meerk40t-ui-manual/full-screens/	All 19 originals from Full screens
images/meerk40t-ui-manual/menu/	All 52 originals from Menu

PDF: Install [Pandoc](#), then run docs/meerk40t/scripts/build-ui-manual-pdf.ps1 → docs/meerk40t/MeerK40t-UI-Manual.pdf.

Image paths in this file use ../../images/meerk40t-ui-manual/ (relative to docs/meerk40t/).

Related docs in this project

Doc	Topic
05-gui-architecture.md	Code-level GUI map
07-planner-and-spooler.md	Cut plans, spooler
11-core-elements-and-io.md	Elements tree, undo
14-console-quickref.md	Console commands
17-meerkat-dlc32-workflow.md	Your DLC32 burn workflow

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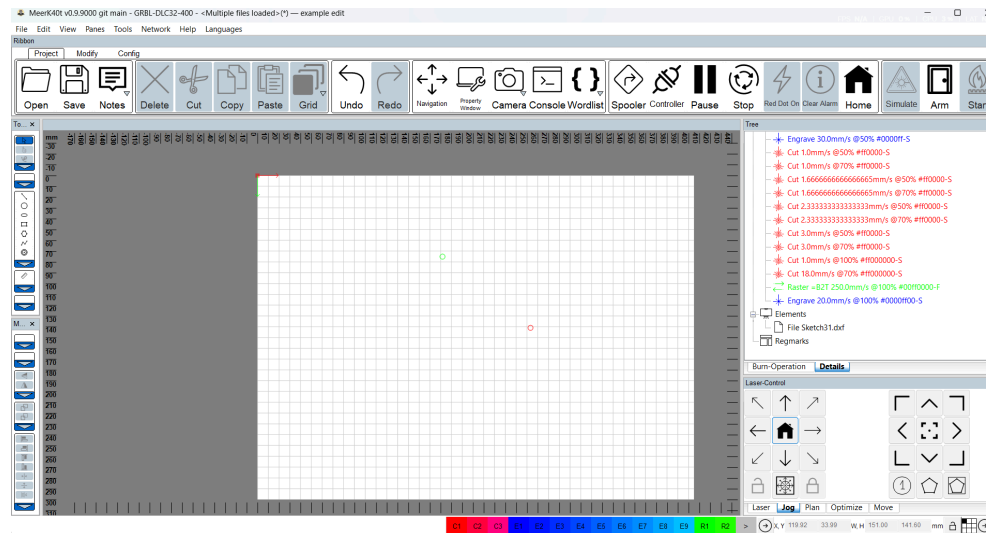
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0. Illustrated walkthrough (your screenshots)

Your session used project **example edit** with **Sketch31.dxf** (MKS DLC32 screen cover) loaded. Below maps **your PNGs** to what each screen does.

0.1 Main window — everything at a glance



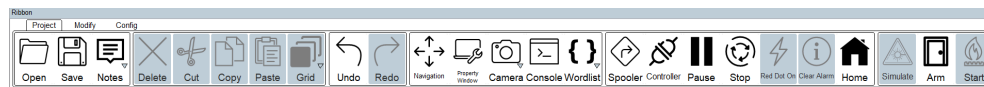
MeerK40t main window — v0.9.9000, GRBL-DLC32-400

Area	What you see	What it does
Title bar	MeerK40t v0.9.9000 · GRBL-DLC32-400 · example edit	Active device profile and project name

Area	What you see	What it does
Menu bar	File, Edit, View, Panes, Tools, Network, Help, Languages	All top-level commands
Ribbon → Project	Open, Save, Undo, Navigation, Console, Spooler, Home, Arm, Start	Daily shortcuts (see §0.2)
Ribbon → Modify	Group, Ungroup, Path Combine, Path Break Apart , booleans, flip, rotate	Unmerge = Path Break Apart (after Combine)
Left toolbars	Draw shapes + align/distribute	Create and lay out art
Scene (center)	Grid, mm rulers, red/green origin	Bed preview; click to select
Tree (right)	Operations · Elements · Regmarks	Job logic + file list
Laser-Control	Jog tab, arrows, Home, keypad	Move head without burning
Color bar (bottom)	C1–C3, E1–E9, R1–R2	Quick color → operation mapping

Typical burn path: classify shapes → **Queue** (Laser/Plan tab or Spooler) → **Simulate** → **Connect** → **Home** → **Arm** → **Start**.

0.2 Ribbon — Project tab



Ribbon Project tab

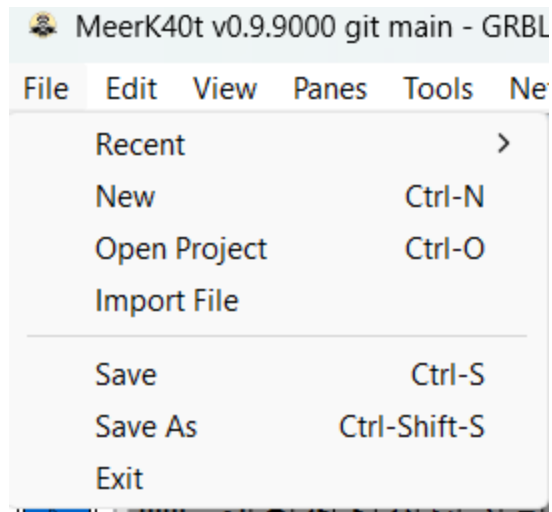
Button	When to use
Open / Save / Notes	Project file (SVG) and job notes
Undo / Redo	After merge, move, delete — Ctrl+Z undoes Merge elements
Navigation	Opens full Navigation window (§0.8)
Console	Type \$101=172, \$HY, \$\$, plan... spool

Button	When to use
Spooler	See queued jobs (§0.10)
Controller	Live GRBL status
Pause / Stop	During a job
Clear Alarm	After GRBL ALARM — then fix cause, often \$X in console
Home	Homing cycle (your profile: sequential <i>HY * *then * *HX</i>)
Simulate	Path preview before burning (§0.11)
Arm	Software enable before fire
Start	Run spooled job (grey until ready)

Modify tab (on main window shot): **Path Break Apart** splits combined paths; **Ungroup** splits groups only.

0.3 Menus you captured

File



File menu

- **Import File** — how **Sketch31.dxf** entered the project (keep **Open Project** for full .svg saves).

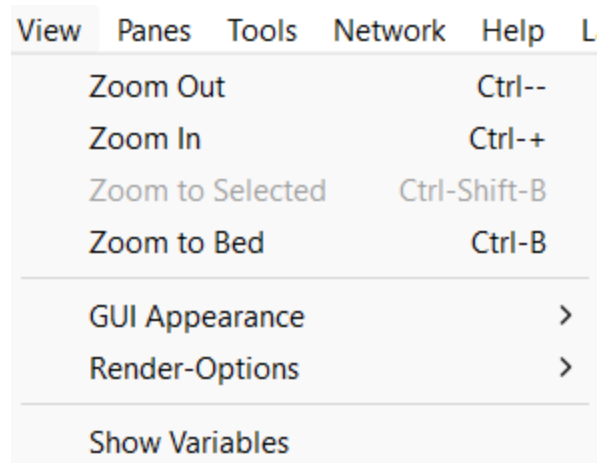
Edit

Edit	View	Panes	Tools	Network	Help
Undo					Ctrl-Z
Redo					Ctrl-Shift-Z
Editing History					>
Cut					Ctrl-X
Copy					Ctrl-C
Paste					Ctrl-V
Paste from system					Shift-Ctrl-V
Select all					Ctrl-A
Select None					
Invert selection					Ctrl-I
Select all non-assigned					
Delete					
Properties					
Device-Manager					
Device-Configuration					
Settings					>

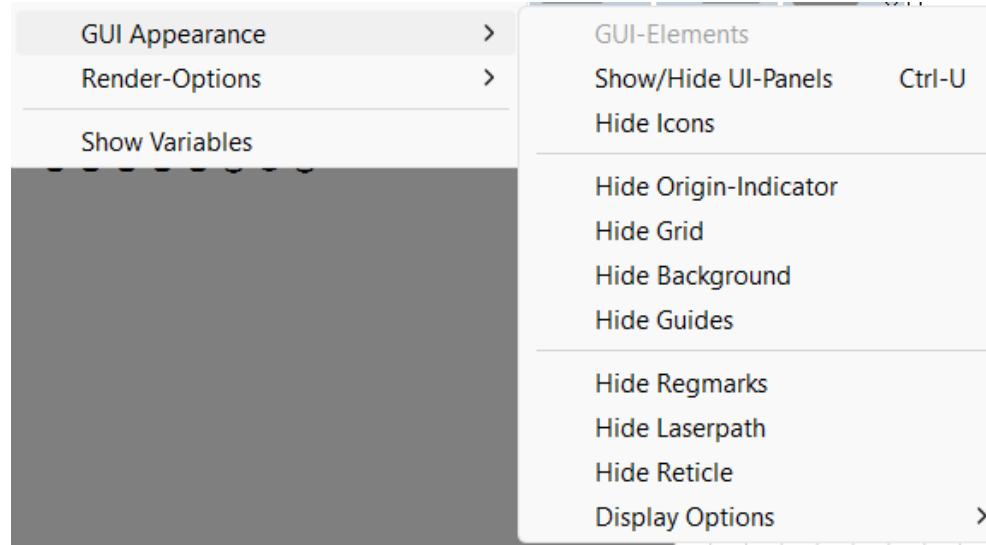
Edit menu

- **Device-Manager / Device-Configuration** — profiles and **GRBL-DLC32-400** settings.
- **Select all non-assigned** — finds shapes that will **not** burn.

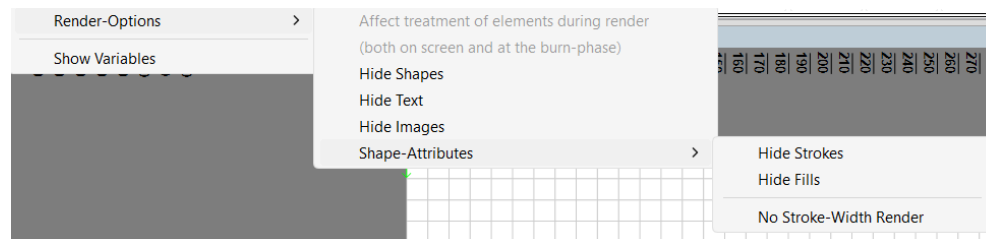
View → GUI Appearance / Render-Options



View menu



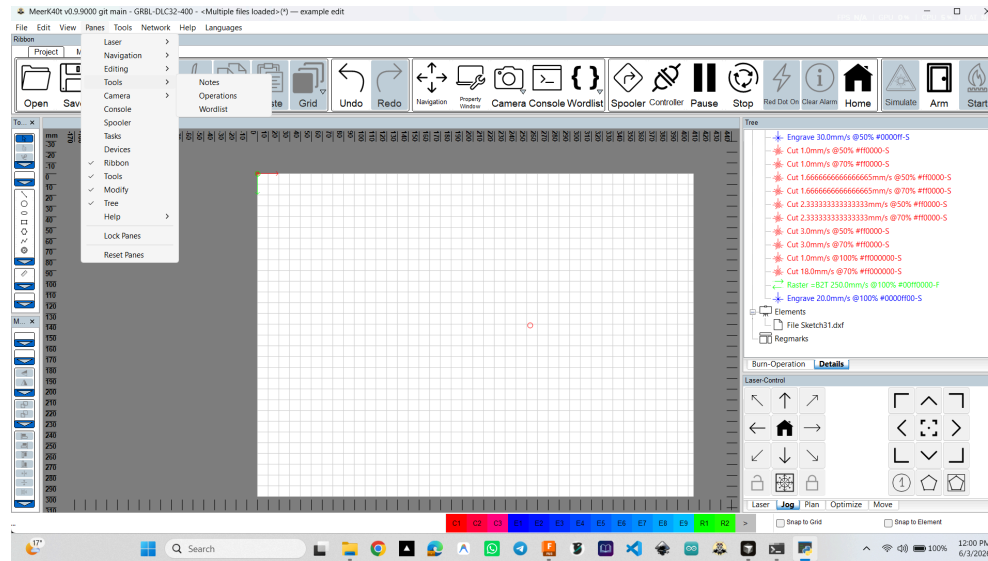
GUI Appearance



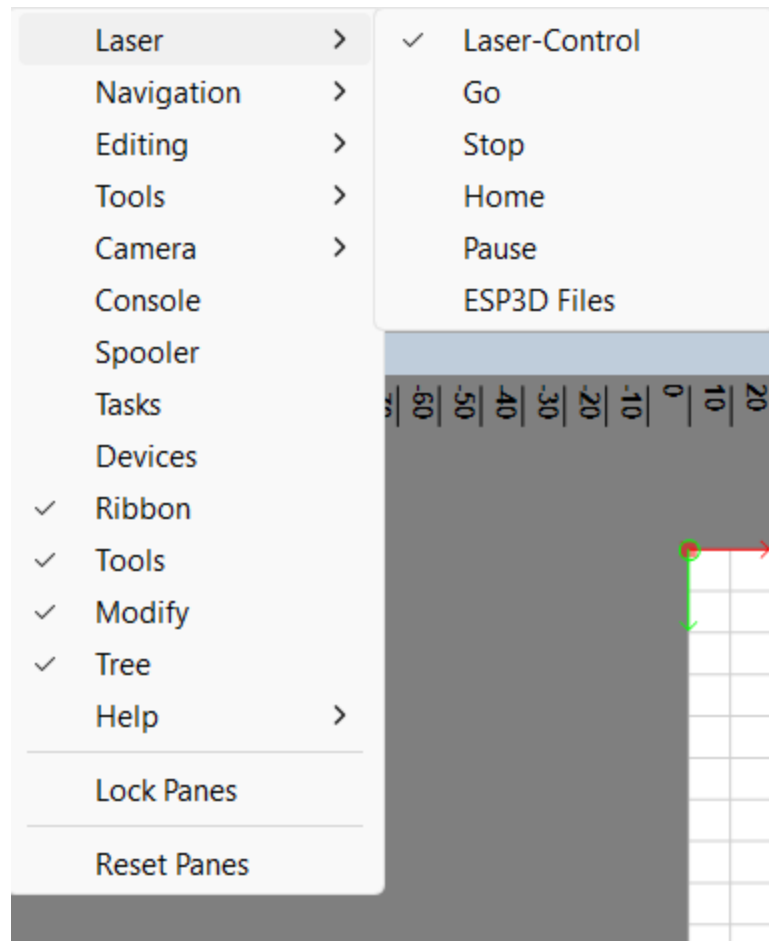
Render-Options

- **Ctrl+U** — show/hide all panels.
- **Hide laserpath / grid / reticle** — troubleshooting display only.
- **Render-Options** can affect burn preview — change carefully.

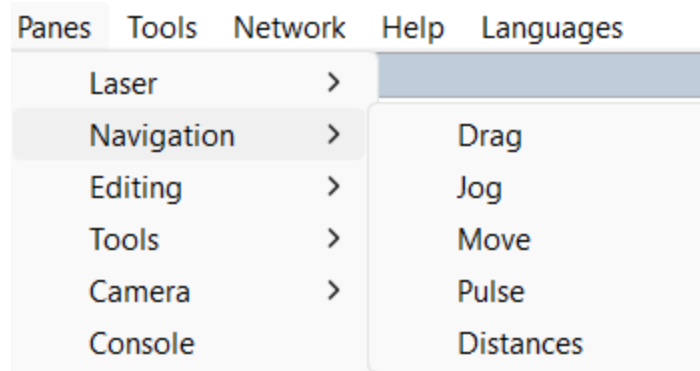
Panes



Panes menu overview



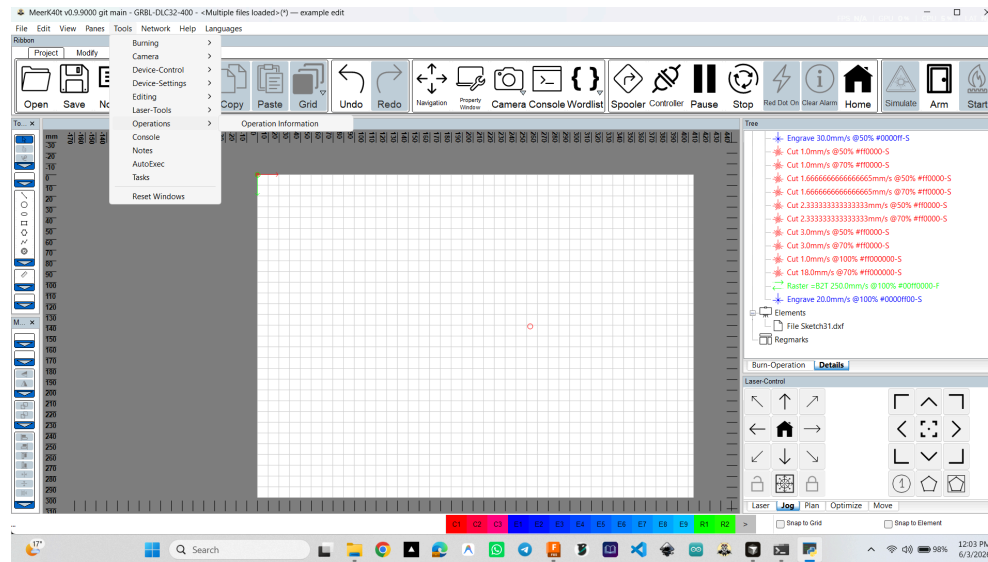
Panels → Laser submenu



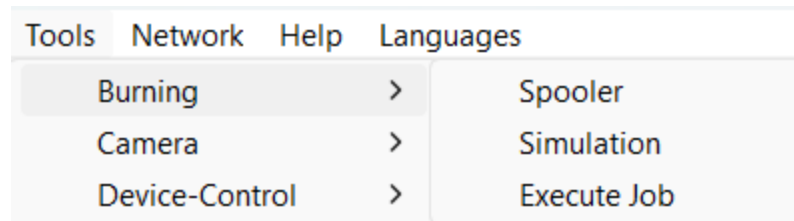
Panels → Navigation submenu

- Checked: **Ribbon, Tools, Modify, Tree, Laser-Control.**
- **Reset Panes** if a window disappears off-screen.

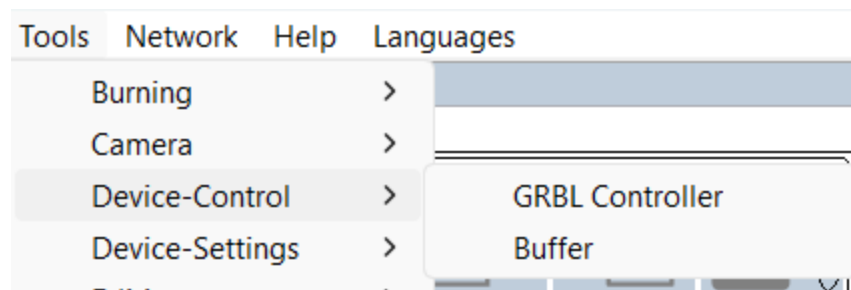
Tools



Tools menu



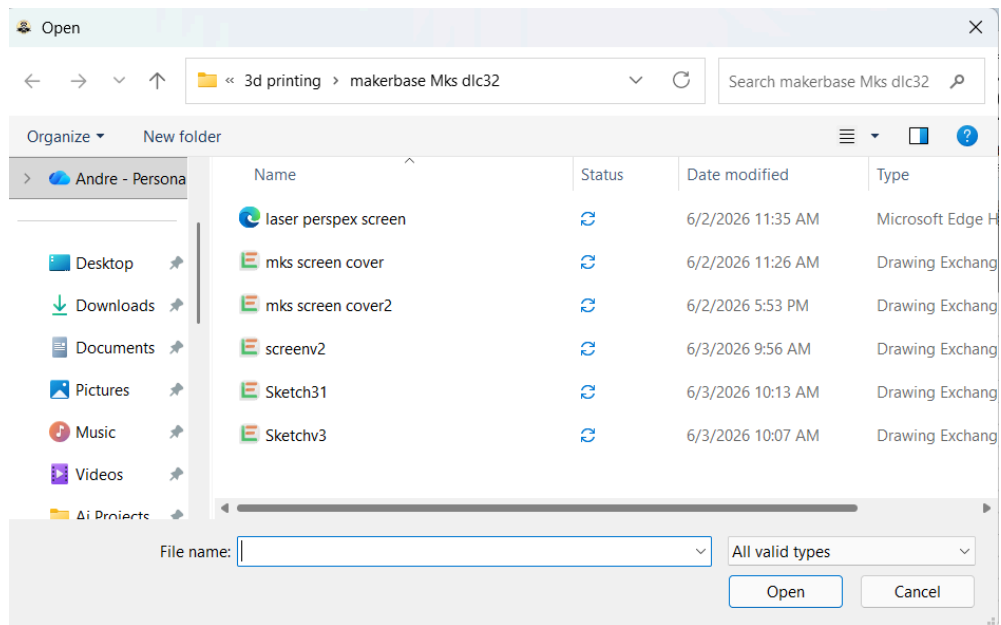
Tools → Burning



Tools → Device-Control

Submenu	Opens
Burning	Spooler, Simulation, Execute Job
Device-Control	GRBL Controller, Buffer (lines queued to board)
Editing	Parameter-Test, Kerf-Test, Alignment, templates, ...
Camera	Webcam overlay (optional)

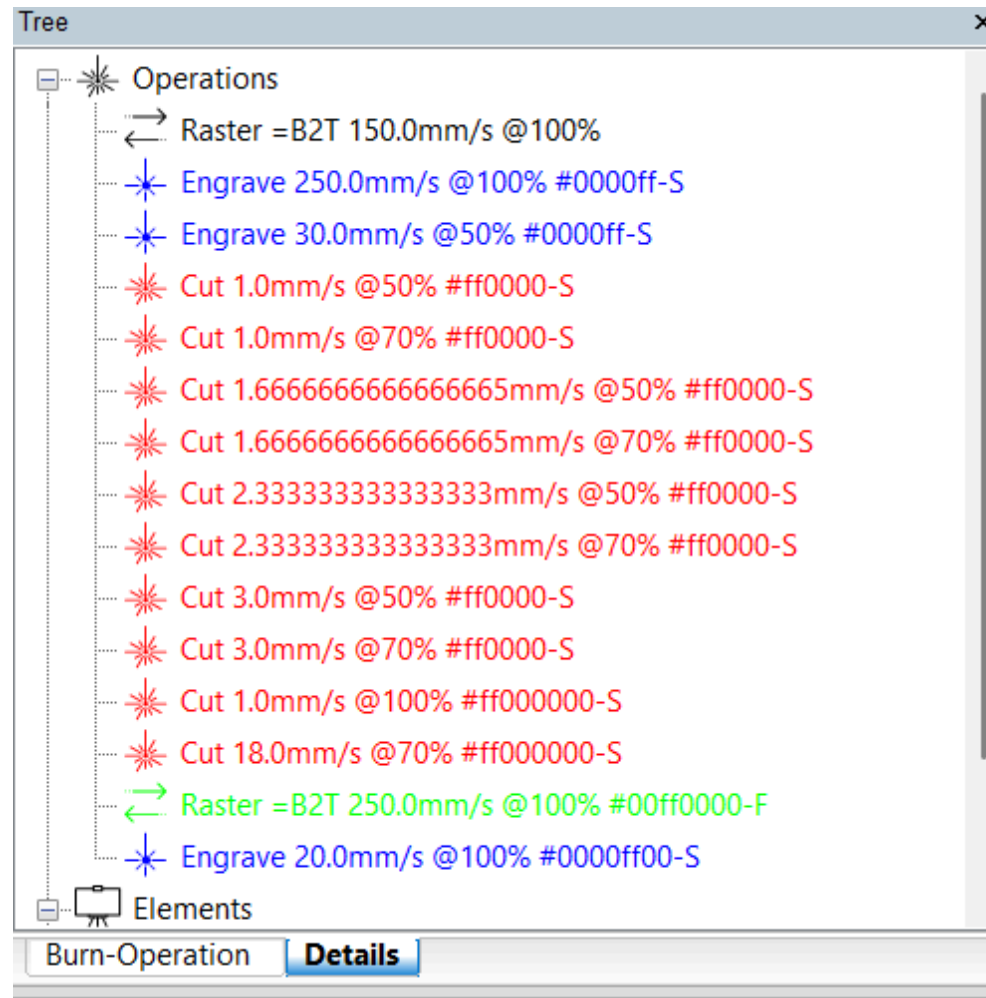
0.4 Import artwork (DXF)



Windows Open — import DXF

You browsed makerbase Mks dlc32 and selected **Sketch31.dxf**. After import, the file appears under **Elements** → **File Sketch31.dxf** in the Tree.

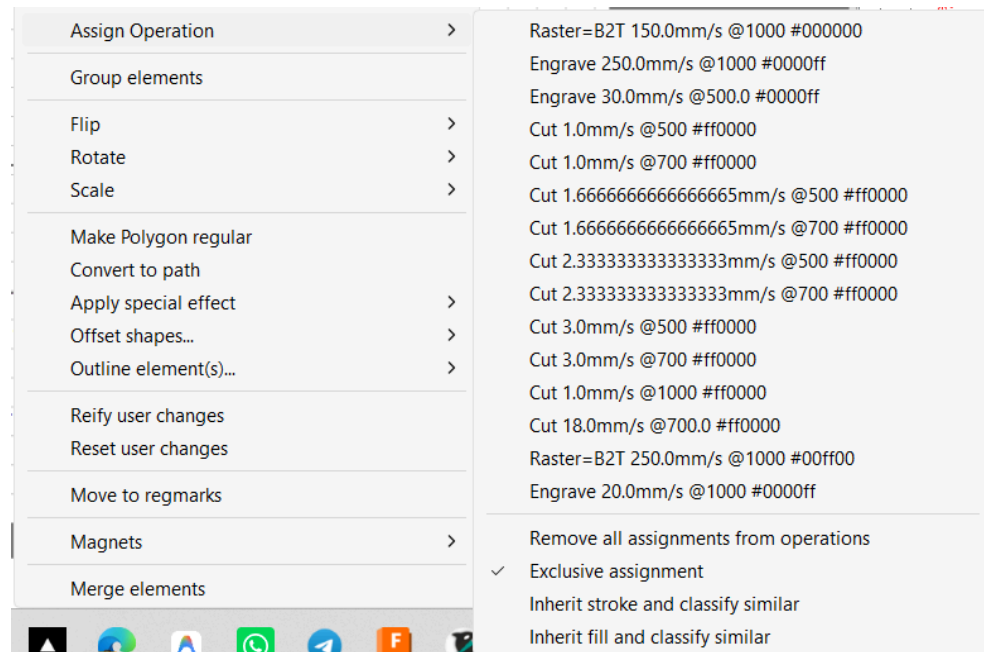
0.5 Operations tree — what will burn



Operations tree

- **Operations** run top to bottom (before global optimize).
- Your list includes many **Cut** speeds (1–18 mm/s) and **Engrave** / **Raster** lines.
- Label format: Cut 3.0mm/s @ 50% #ff0000-S = speed, power %, colour, stroke (S) vs fill (F).
- **Elements** holds the DXF; expand to select individual paths.

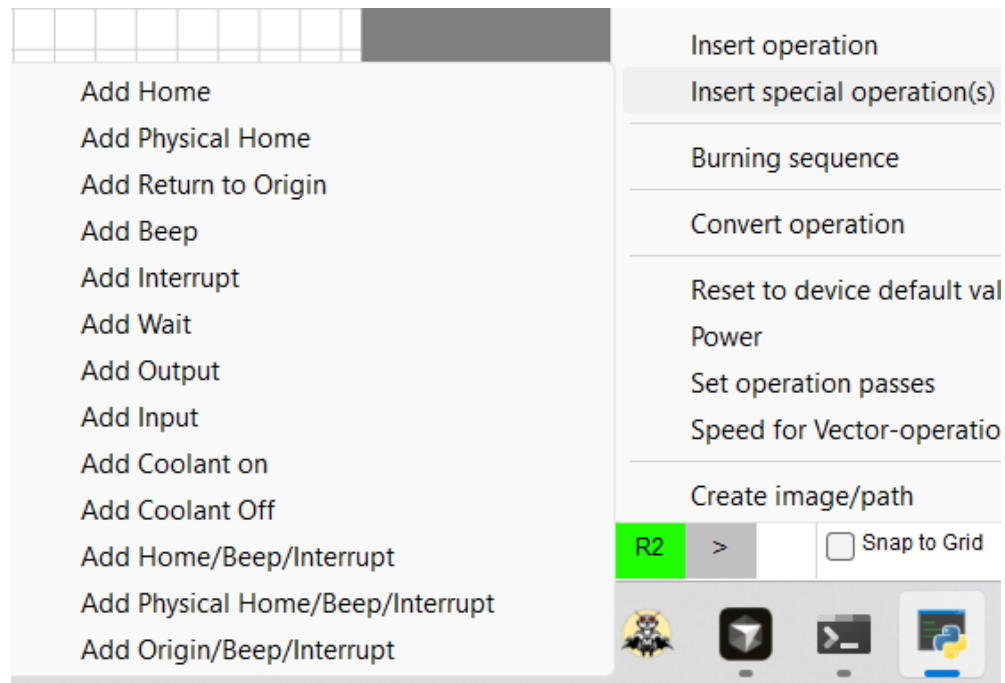
Assign Operation (right-click shape):



Assign Operation submenu

- Pick the Cut/Engrave line that matches your material test.
- **Exclusive assignment** ✓ — one operation per shape.
- **Inherit stroke/fill and classify similar** — auto-assign by colour.

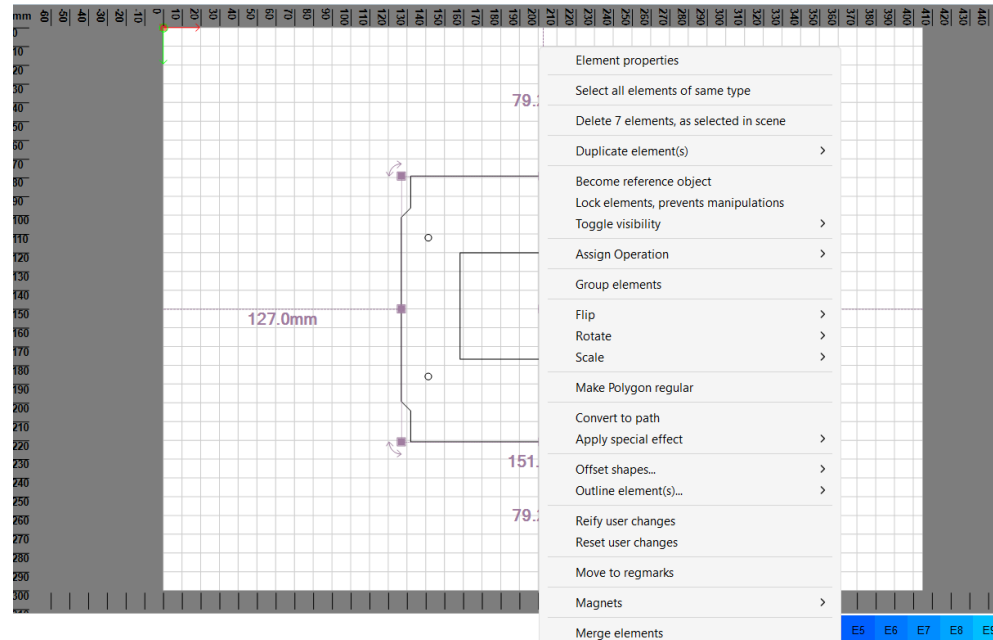
Operations right-click — insert homing/wait:



Insert special operations

- **Add Physical Home** — use between passes if needed (you also have **Physical Home** at job end in config).
-

0.6 Scene — select, measure, context menu



Scene context menu — 7 elements selected

Command	Purpose
Element properties	Stroke, fill, size
Assign Operation	Link to Cut/Engrave
Group / Merge elements	Combine selection
Convert to path	Editable nodes
Reify / Reset user changes	Apply or undo scale-rotate

Purple handles = resize/rotate. Dimension labels (e.g. **127.0 mm**) show selection size on bed.

0.7 GRBL device configuration (your DLC32 profile)

Device tab — bed size and axis flip:

GRBL-Configuration

Device Interface Protocol Advanced Effects Operation Defaults ESP3D Upload Warning Default Actions Display

Label
GRBL-DLC32-400

Dimensions

Width 405.0mm Height 300.0mm Laserspot 0.3mm

Scale

X-Axis 1.0 Y-Axis 1.0

Flip Axis

☐ Flip X ☒ Flip Y

Axis corrections

☐ Swap XY

Home position

Force Declared Home
top-left

Z-Axis support

☐ Supports Z-axis Z-Homing \$HZ

Screen updates

☒ Device Position

Auto-Configuration

Query properties
Hardware properties

GRBL Configuration — Device

Setting	Your value	Note
Label	GRBL-DLC32-400	Matches title bar
Width × Height	405 × 300 mm	Match \$130 / \$131 on board
Flip Y	✓	Matches machine (into bed = Y-)
Force home	top-left	After homing, origin top-left

Advanced tab — homing and buffer (critical):

GRBL-Configuration

Device Interface Protocol Advanced Effects Operation Defaults ESP3D Upload Warning Default Actions Display

Red Dot

☐ Simulate reddenot

Reddot Laser strength /1000

If active then you can turn on the laser at a very low power to get a visual representation of the current position to help with focusing and positioning. Use with care!

☒ Active during Outline

Maximum speeds

Max vector speed mm/s

Max raster speed mm/s

Travel

Travel speed mm/s

Controller Buffer

☒ Limit the controller buffer size

Controller Buffer lines

Config

☐ Use M3

☐ Use G1 before Rapid Moves

☐ Reset on 'Clear Alarm'

Curve Interpolation

☒ Device has endstops

☒ Sequential homing (\$HY then \$HX)

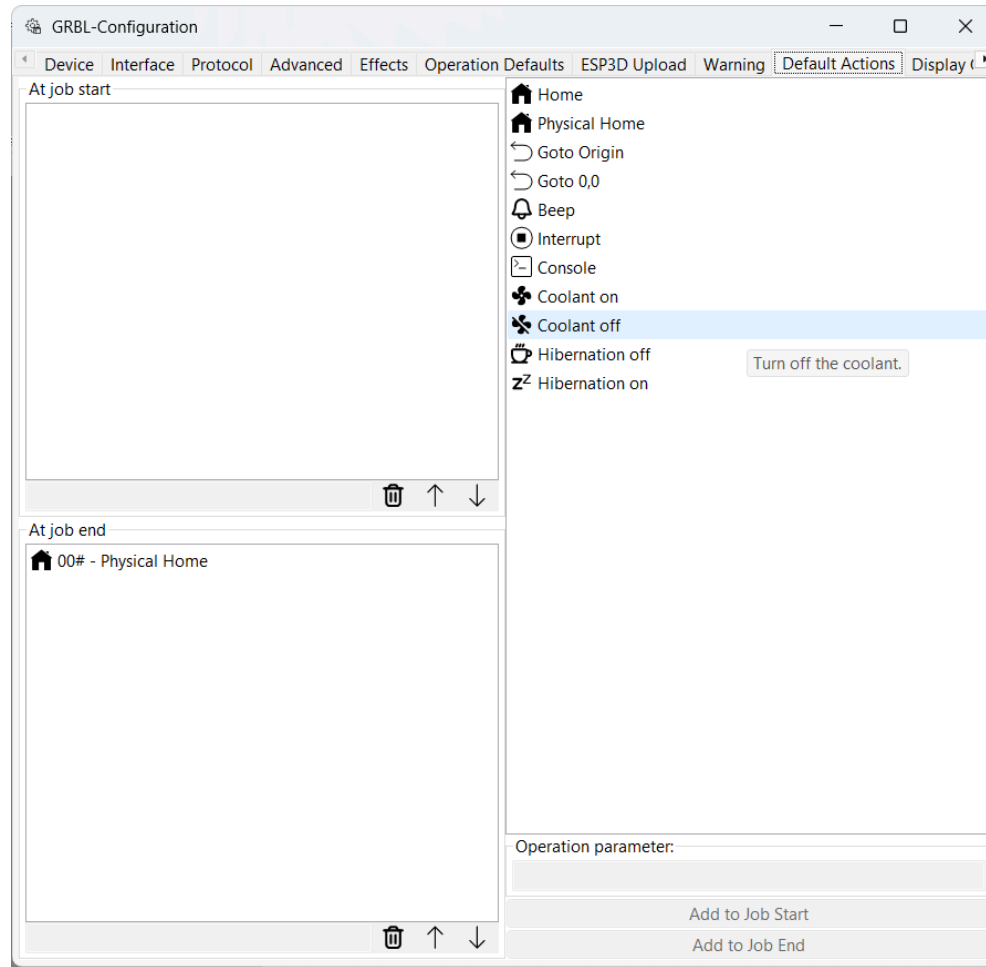
Coolant Support

Coolant

GRBL Configuration — Advanced

Setting	Your value	Note
Device has endstops	✓	Required for your limit switches
Sequential homing (\$HY then <i>HX</i>) * * ✓ <i>Avoidsingle</i> * *H on your machine		
Controller buffer	200 lines	WiFi/USB stability

Default Actions — after every job:



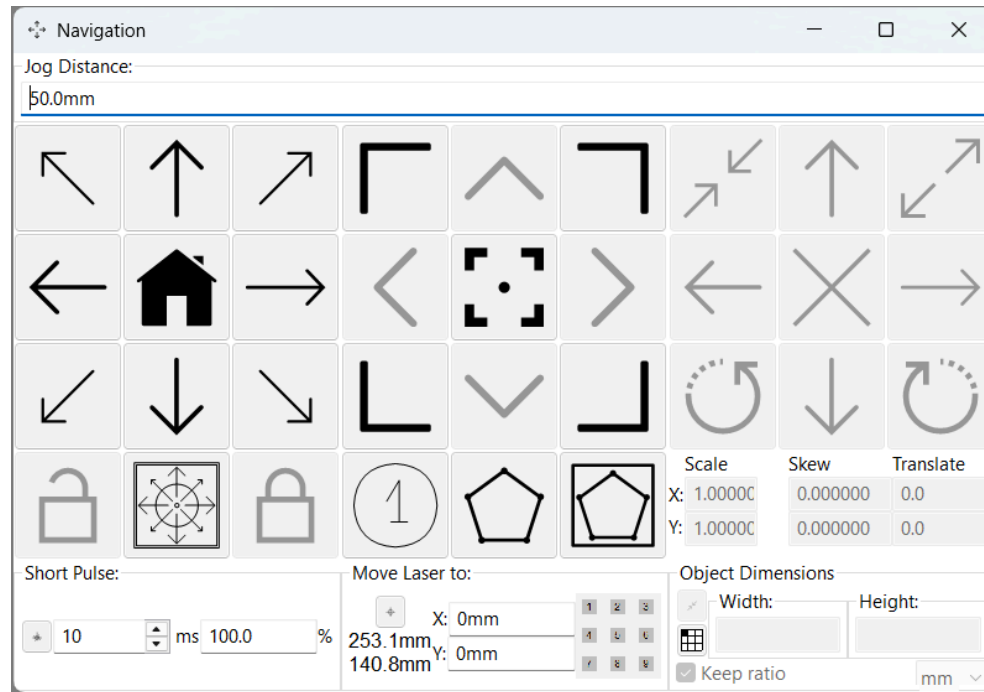
GRBL Configuration — Default Actions

- **At job end:** Physical Home — head returns to switches.

Other tabs captured: Effects, Display Options — see 11c-grbl-config-effects.png, 11e-grbl-config-display-options.png in image folder.

Steps/mm calibration (**\$100**, **\$101**) is done in the **Console**, not on these tabs (Device **Scale** stays **1.0**).

0.8 Navigation window — jog, pulse, move



Navigation window

Section	Use
Jog distance	e.g. 50 mm for calibration moves
Arrow pad + Home	Manual jog; centre = home
Short Pulse	Brief beam for alignment (chiller + WP must be OK)
Move laser to	Type X/Y (your shot: ~253.1 / 140.8 mm)
Scale / translate	Numeric transform on selection

Same jog controls exist in docked **Laser-Control** → **Jog** on the main window.

0.9 Material tuning tools

Parameter-Test — speed vs power grid:

Parameter-Test

Operation to test
Operation: **Cut**

Show Labels / Values
☒ Labels ☒ Values ☒ Simple

First parameter (X-Axis)
Subtitle:
Parameter: **Speed**
Count: **4** Every 0.667mm/s
Minimum: **1** mm/s
Maximum: **3** mm/s
Width: **10** mm
Delta: **3** mm
Color: **Red** ☐ Growing

Second parameter (Y-Axis)
Subtitle:
Parameter: **Power**
Count: **2** Every 400%
Minimum: **300** %
Maximum: **700** %
Height: **10** mm
Delta: **5** mm
Color: **Blue** ☐ Growing

Create Pattern

Queue to Spooler

How to use it

To provide the best burning results, the parameters of operations need to be adjusted according to *YOUR* laser and the specific material you want to work with (e.g. one batch of poplar plywood from one supplier may respond completely different to a batch of another supplier despite having the very same specifications on paper).
E.g. for a regular CO2 laser you want to optimize the burn speed for a given power to reduce burn marks or decrease execution time.

Generator Properties Advanced Templates

Parameter-Test

1. Set operation **Cut**, X = speed, Y = power.
2. **Create Pattern** on bed.
3. Burn → pick best square → apply to real Cut operation.
4. **Queue to Spooler** sends grid to machine.

Kerf-Test — fit test pieces for beam width:

 Kerf-Test

—

□

×

Cut-Operation

Speed

mm/s

Power

%

Parameters

Pattern

☒ Rectangular (box joints) ☐ Circular (inlays) ☐ Slider

Count:

Every 0.05mm

Minimum:

Maximum:

Size:

Delta:

 Create Pattern

How to use it



If you want to produce cut out shapes with **exact** dimensions after the burn, then you need to take half the width of the laserbeam into consideration (aka Kerf compensation).
This routine will create a couple of test shapes for you to establish this value.
After you cut these shapes out you need to try to fit the shapes with the same label together. Choose the pair that has a perfect fit and use the label as your kerf-compensation value.

Kerf-Test

0.10 Job Spooler

The screenshot shows a window titled "Job Spooler" with standard window controls. Below the title bar is a toolbar with a dropdown menu showing "-- All available devices --", a "Select the device" button, a "Pause" button with a pause icon, and a red "Abort" button with a stop icon. The main area contains two tables. The top table is for active jobs with columns: #, Device, Name, Item, Status, Type, Steps, Passes, Priority, and Runtime. The bottom table is for completed jobs with columns: #, Device, Name, Start, End, Runtime, Estimate, Steps, Passes, and Status. A "Clear History" button is located to the right of the completed jobs table header.

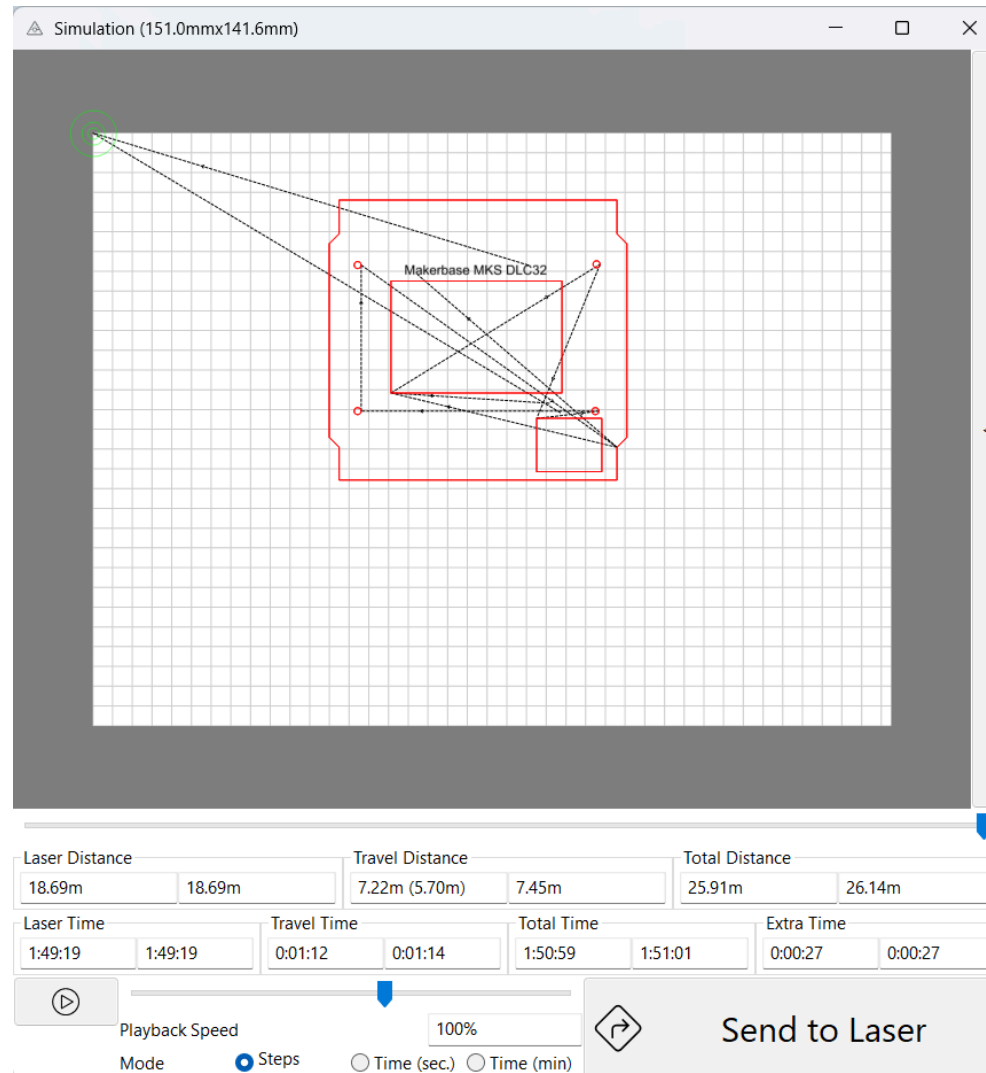
#	Device	Name	Item	Status	Type	Steps	Passes	Priority	Runtime
---	--------	------	------	--------	------	-------	--------	----------	---------

#	Device	Name	Start	End	Runtime	Estimate	Steps	Passes	Status
---	--------	------	-------	-----	---------	----------	-------	--------	--------

Job Spooler

- **Top:** active queue — **Pause**, **Abort**.
 - **Bottom:** completed jobs history.
 - Empty tables = nothing queued yet (use **Queue** on Laser pane first).
-

0.11 Simulation before Start



Simulation — Sketch31 / MKS cover

- **Red** = cutting; **dashed black** = travel (laser off).
- Shows **laser time**, **travel time**, **total distance**.
- Your job example: ~1 h 51 min total — use to decide if order is acceptable.
- **Send to Laser** when satisfied (still need **Arm** + **Start** on main UI).

1. Big picture: how MeerK40t is organized

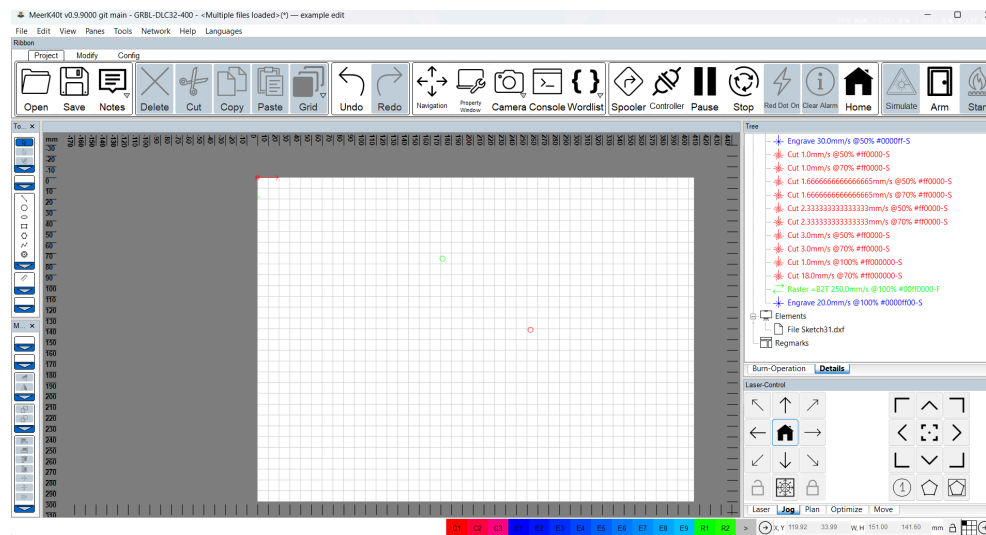
MeerK40t is built around a **kernel** (command processor) with a **graphical shell** on top.

Menus • Ribbon • Status bar • Job buttons (Go/Stop/Home)		
Tree panes Elements Operations	SCENE (draw, select, preview)	Laser / Nav (device)
Console (commands, GRBL \$ settings, errors)		

Three ideas to remember:

1. **Elements** = what you drew or imported (geometry, text, images).
2. **Operations** = how the laser treats those elements (cut, engrave, raster, etc.).
3. **Device + Spooler** = connection to the controller and the queued job that actually runs.

Nothing burns until: shapes are **classified** into operations → plan is **built** → job is **spooled** → you **arm** and **start** (GRBL: connect first).



Main window

2. Starting the application

Normal GUI

- Run MeerK40t from your install or `python -m meerk40t` from the clone.
- A **splash / busy** screen may appear while plugins load (`gui/plugin.py`).

Simple UI (`-w / --simpleui`)

A reduced interface without the full main window. Use the full GUI for learning and for CO2 workflow.

Persistence

- **Project** = saved as **SVG** (File → Save). Operations and elements live in that file.
 - **Window layout** can be remembered (Window menu → reset option clears stored pane positions).
 - **Device settings** live in MeerK40t config + often mirrored on the board (GRBL \$\$).
-

3. Main window layout

The main frame is class **MeerK40t** (`gui/wxmmain.py`). Layout uses **wxAUI** — panes dock, float, tab, and hide.

Typical panes (defaults vary by profile)

Pane	Role
Scene	Main bed view; draw and select here
Ribbon	Tool buttons (line, rect, edit nodes, ...)
Tree	Elements + Operations tabs
Laser	Device, connect, arm, start, plan/spool

Pane	Role
Console	Type commands; see GRBL traffic
Position	Cursor / selection coordinates
Go / Stop / Pause / Home	Quick job and homing buttons (often in toolbar)

Open or hide panes: **View** menu and **Pane** submenu (dynamic list of registered wxpane/* panels).

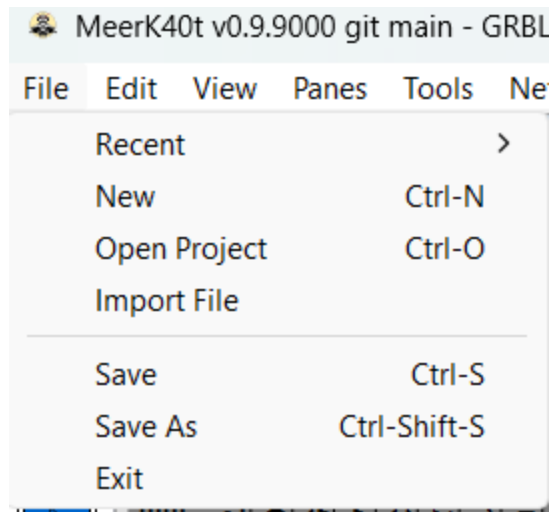
Title bar

Shows project name, device name, and status hints (connection, alarms). Exact format depends on version and connection state.

4. Menus

4.1 File

Item	Shortcut	What it does
New	Ctrl+N	Clears operations, elements, notes (fresh project)
Open Project	Ctrl+O	Replaces project from SVG; Shift = pick folder
Import File	—	Adds file into current project (SVG, DXF, images, ...)
Save	Ctrl+S	Overwrite current SVG project
Save As	Ctrl+Shift+S	Save under new name
Recent	—	Recent projects (not on some frozen macOS builds)
Exit	—	Quit application



File menu

Tip: Your machine workflow often uses **Import** for artwork and **Save** before every long burn session.

4.2 Edit

Item	Shortcut	What it does
Undo	Ctrl+Z	Reverses last change (merge, move, delete, ...)
Redo	Ctrl+Shift+Z	Restores undone change
Editing History	—	Submenu to jump to an earlier undo state
Cut / Copy / Paste	Ctrl+X/C/V	Clipboard inside MeerK40t
Paste from system	Shift+Ctrl+V	Paste from other apps (e.g. Illustrator paths)
Select all	Ctrl+A	Emphasize every element
Select None	—	Clear selection
Invert selection	Ctrl+I	Toggle selection

Item	Shortcut	What it does
Select all non-assigned	—	Highlights shapes not in any operation
Delete	—	Remove selected tree items
Properties	—	Opens Properties window for selection
Device-Manager	—	Add/switch laser devices
Device-Configuration	—	Active device settings (GRBL port, speeds, ...)
Preferences	—	Global GUI and behavior
Wordlist	—	Variable text lists
Hershey Font Manager	—	Stroke fonts for engraving
Keymap	—	Keyboard shortcut editor

Edit	View	Panes	Tools	Network	Help
Undo					Ctrl-Z
Redo					Ctrl-Shift-Z
Editing History					>
Cut					Ctrl-X
Copy					Ctrl-C
Paste					Ctrl-V
Paste from system					Shift-Ctrl-V
Select all					Ctrl-A
Select None					
Invert selection					Ctrl-I
Select all non-assigned					
Delete					
Properties					
Device-Manager					
Device-Configuration					
Settings					>

Edit menu

Merge / unmerge: There is no menu item named “Unmerge”. Use **Undo** after **Merge elements**, or right-click a path → **Break Subpaths** (see §8).

4.3 View

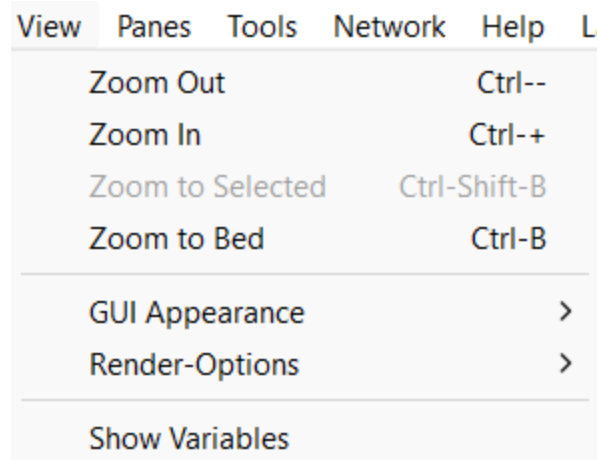
Controls **what is drawn** on the scene and in the burn preview — not the laser hardware.

GUI Appearance (examples):

- Theme / colors
- Icon sizes
- Tooltip and overlay behavior

Scene Appearance (examples):

- **Hide Shapes / Text / Images** — toggles draw categories
- **Render-Options** — advanced: affects on-screen and burn-phase treatment (use carefully)
- **Show origin**, grid, guides, magnet targets
- **Show variable replacement** — preview wordlist substitutions in text



View menu

If the bed looks “empty” but the tree has items, check **Hide Shapes** and emphasis (selection).

4.4 Pane menu

Lists all registered dock panels (Console, Tree, Laser, Navigation sub-panes, Snap options, etc.). Check = visible.

4.5 Tool / Window menu (dynamic)

Window menu is built from registered window/* modules. Grouped entries open tools such as:

- Alignment, Simulation, Material Manager, Split Image, Kerf, Living Hinge, ...

Reset window positions — restores default layout when panes are lost off-screen.

4.6 Network

Service	Purpose
Webserver	Control MeerK40t from a browser (port in settings)
Telnet / consoleserver	Remote console
GRBL-server	Third-party apps talking GRBL through MeerK40t
Ruida-server	Same idea for Ruida

*Network menu — not captured in this session; use menu bar **Network** for Webserver / Telnet / GRBL-server.*

4.7 Help

- Online help (context-sensitive where implemented)
 - Offline help bundle (if shipped with your build)
 - About, Tips window, links to wiki/GitHub
-

5. Dockable panes

Registered in `gui/wxmeerk40t.py` and related modules.

Registration	User-facing role
<code>wxpane/ScenePane</code>	Scene container

Registration	User-facing role
wxpane/Ribbon	Tool ribbon
wxpane/Tree	Elements + Operations
wxpane/LaserPanel	Device + burn controls
wxpane/Navigation	Jog, pulse, transforms (sub-panes)
wxpane/Position	Live coordinates
wxpane/opassign	Quick operation assignment
wxpane/Snap	Snap distances and modes
wxpane/magnet	Magnetic alignment helpers
wxpane/wordlist	Wordlist quick access
pane/console	Console
pane/laser	Alternate laser strip
pane/jog, pane/drag, pane/move, pane/pulse, pane/transform	Navigation pieces

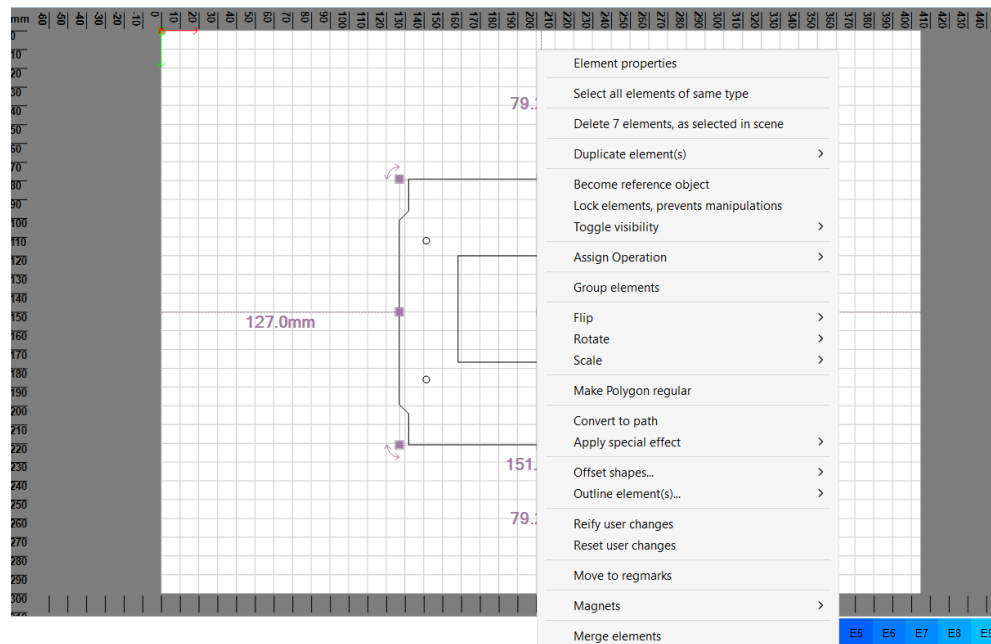
Drag pane title bars to dock left/right/bottom or float. Right-click pane tab for hide/close.

6. Scene (work area)

The scene (gui/wxmscene.py, gui/scene/scene.py) is your **virtual bed**.

Navigation in the scene

Action	Typical input
Pan	Middle-mouse drag or dedicated pan mode
Zoom	Mouse wheel or zoom tool
Select	Click shape; shift-click add; drag rectangle
Move selection	Drag selected objects
Resize / rotate	Handles when selection tool active



Scene — selection and context menu

Emphasis vs selection

- **Emphasized** items (highlighted in tree) are what operations and Properties act on.
- Many menu items require `has_emphasis` (something selected).

Bed size and origin

Comes from **device settings** (for GRBL: workspace size, origin corner). Scene grid should match 130/131 soft limits after calibration.

Colors

Stroke/fill in scene reflect element attributes; operation colors may show in preview modes. **GuiColors** service drives themes.

7. Drawing and editing tools

Tools register under tool/* in wxmscene.py.

Tool ID	Name (typical)	Use
tool/draw	Draw	Freehand paths
tool/rect	Rectangle	Boxes
tool/line	Line	Single segments
tool/polyline	Polyline	Connected segments
tool/polygon	Polygon	Closed polygon
tool/circle	Circle	Circles
tool/ellipse	Ellipse	Ellipses
tool/point	Point	Mark points
tool/text	Text	Text boxes
tool/linetext	Line text	Text along a line
tool/vector	Vector	Vector-specific ops
tool/measure	Measure	Distance measurement
tool/relocate	Relocate	Move job origin
tool/placement	Placement	Place imported art
tool/edit	Node Edit	Edit path points
tool/nodemove	Node Move	Move single nodes
tool/pointmove	Point Move	Fine point drag
tool/parameter	Parameter	Edit path parameters
tool/imagecut	Image cut	Cut contours from image
tool/tabedit	Tab edit	Tabs for mechanical parts
tool/ribbon	Ribbon	Meta-tool on ribbon bar



Ribbon — Project tab

Node Edit tool (important)

- Select a **path** → activate **Node Edit**.
- Click points; drag handles for Bézier control.
- **Break** (b key in tool): split path at selected point(s) — different from **Break Subpaths** in tree.
- **Insert** / **Delete** points as needed for cleanup after import.

*Node Edit tool — activate from left toolbar; use **Break** on a point to split a path (see §0.1 Modify ribbon **Path Break Apart** for combined paths).*

8. Elements tree

Left tree tab **Elements** lists all geometry groups.

Structure

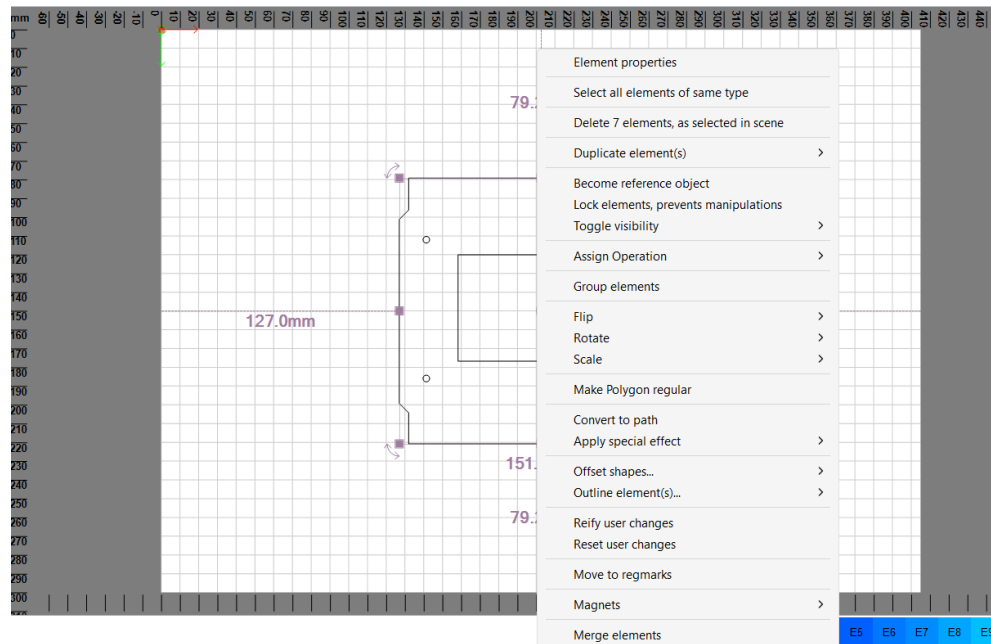
- **Group** — folder-like container
- **elem path**, **elem rect**, **elem image**, **elem text**, ... — leaf types
- **Regmark** / guide layers may appear depending on profile

Right-click (context) actions (common)

Exact list depends on node type and lock state. Examples from `element_treeops.py`:

Action	Effect
Merge elements	Multiple shapes → one path (undo: Ctrl+Z)
Break Subpaths	One path → group of separate paths
Merge items (on group)	Children → single path
Lock / Unlock	Prevent edits
Copy / Paste	Duplicate subtrees
Classify	Auto-assign to operations
Ungroup	Split group

Action	Effect
Reorder (up/down)	Tree order (can affect burn order before planner)



Elements — scene context menu

Merge vs Break Subpaths (recap)

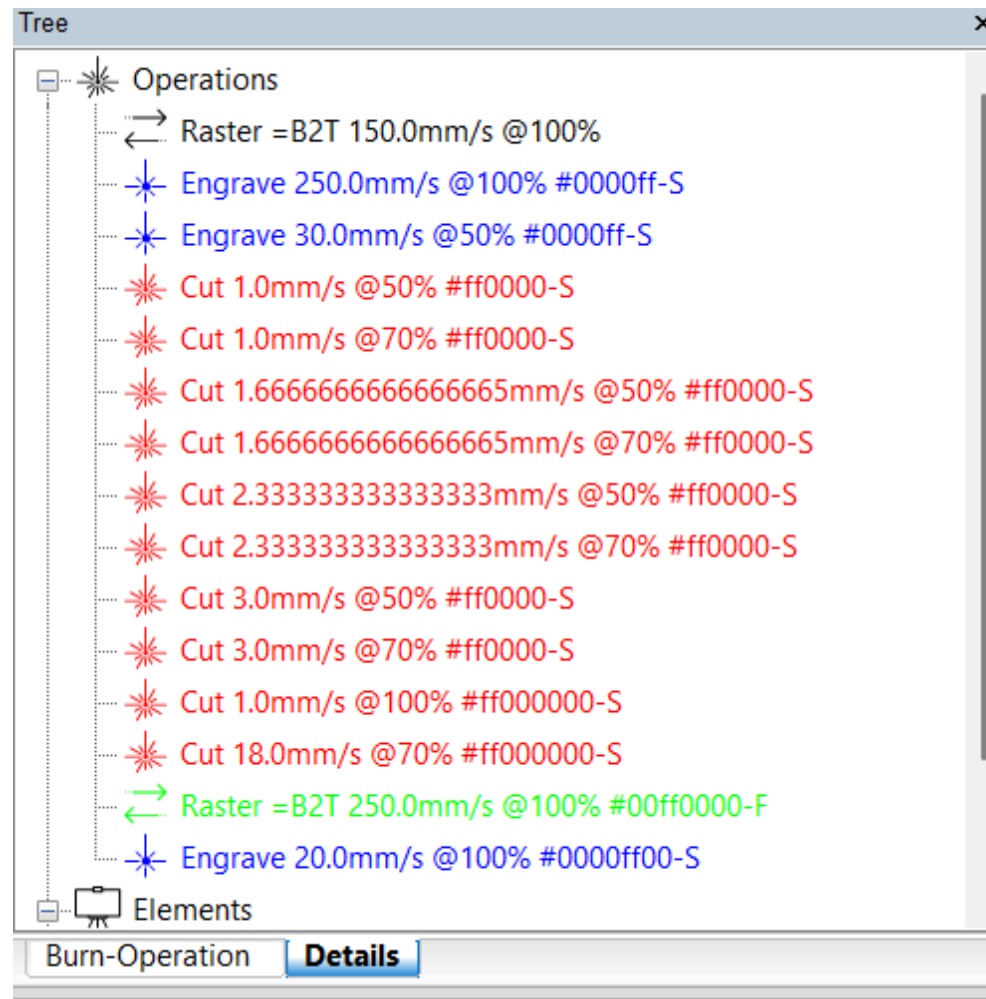
- **Merge** = design choice to combine geometry.
- **Break Subpaths** = splits **disconnected** contours inside one path into separate elements.
- Stitched contours (shared endpoints) may stay one path until you **Break** in node edit or Undo merge.

9. Operations tree and classification

Second tab **Operations** defines **Cut**, **Engrave**, **Raster**, **Image op**, **Dots**, etc.

Concepts

Term	Meaning
Operation	Laser settings + rules for a set of shapes
Classification	Linking elements → operations (stroke color, fill, automatic rules)
Re-Classify	Re-run rules after you change art
Unassigned	Shapes not in any operation — will not burn



Operations tree

Typical CO2 workflow

1. Create **Cut** operation (speed, power, passes).
2. Import or draw art.
3. **Classify** (drag-drop or automatic).
4. Open **Operation Properties** — confirm power/speed, **dot length**, **frequency** (device dependent).
5. If inner holes skip: enable **Fill** on Cut op or assign inner paths to Cut (see [17-meerkat-dlc32-workflow.md](#)).

Planner options (in device/plan config)

- **Merge Operations / Merge Passes** — optimizes order at spool time; does **not** merge shapes in the scene.
-

10. Properties

Window → **Properties** or **Edit** → **Properties**.

Two contexts:

1. **Element properties** — position, size, stroke, fill, font, image DPI, etc.
2. **Operation properties** — power, speed, passes, kerf, dithering, raster direction, GRBL-specific tabs from driver plugins.

*Properties window — open via **Edit** → **Properties** or right-click **Element properties** (not in this screenshot set).*

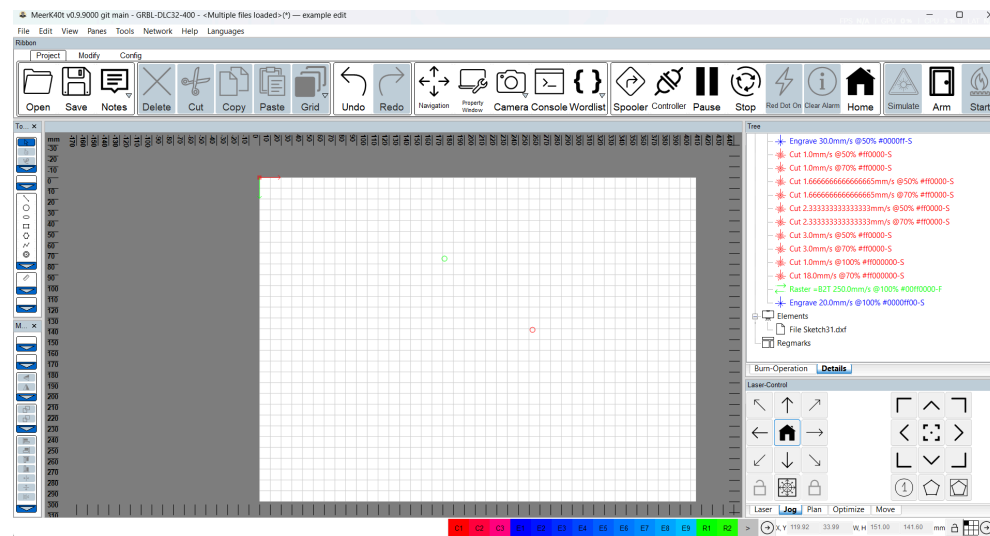
Changes apply to **emphasized** items. Multi-selection may show common fields only.

11. Laser panel and job control

The **Laser** pane (`gui/laserpanel.py`) is the control center for the active device.

Main controls (typical)

Control	Role
Device selector	Which configured laser profile
Connect / Disconnect	Open serial/TCP to controller
Arm	Permit firing (safety interlock in software)
Start / Execute	Run spooled job
Pause / Resume	Pause motion
Stop / E-Stop	Halt job and reset planner state
Queue / Plan	Build cut plan and spool
Hold	Reuse last plan without full rebuild (when blob exists)
Update plan	Refresh plan after edits



Laser-Control pane — bottom right of main window

Safe sequence (GRBL)

1. Chiller / **WP** interlock satisfied.
2. **Connect**.
3. **Home** (\$HY then \$HX on your machine — use Navigation or configured home buttons).

4. **Queue** job (checks unassigned shapes).
5. **Arm** → **Start**.

Unassigned warning

When queueing, MeerK40t may warn that some shapes are not assigned — inner detail may be skipped. Read the dialog; fix via Re-Classify or drag to Cut.

Unassigned-shapes warning — appears when queueing if shapes are not in any operation (see §0.5).

Toolbar job buttons

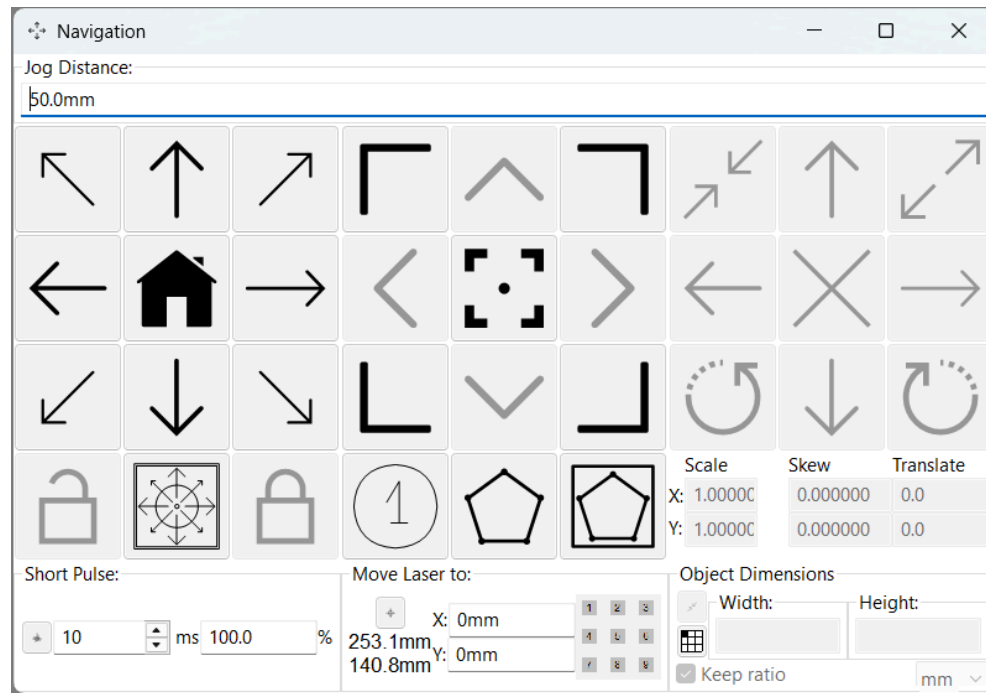
- **Go** — start/spool depending on context
- **Stop** — emergency stop icon
- **Home** — homing cycle
- **Pause** — pause spooler

Colors follow **themes** (arm = caution, stop = red).

12. Navigation and jogging

Window → **Navigation** opens jog controls.

Sub-panel	Function
Jog	Direction buttons, continuous jog
Jog distance	Step size (1 mm, 10 mm, 100 mm, ...)
Move	Go to absolute X/Y
Pulse	Short laser fire for alignment (with caution)
Transform	Scale/rotate selection numerically
Drag	Drag-style positioning



Navigation

GRBL note: Jog moves use controller jog mode; calibrate \$100/\$101 using real measurements (see [16-mks-dlc32-board.md](#)).

Confined coordinates: MeerK40t may limit jog to bed — if jog “stops early”, check soft limits and homing.

13. Planner, spooler, simulation

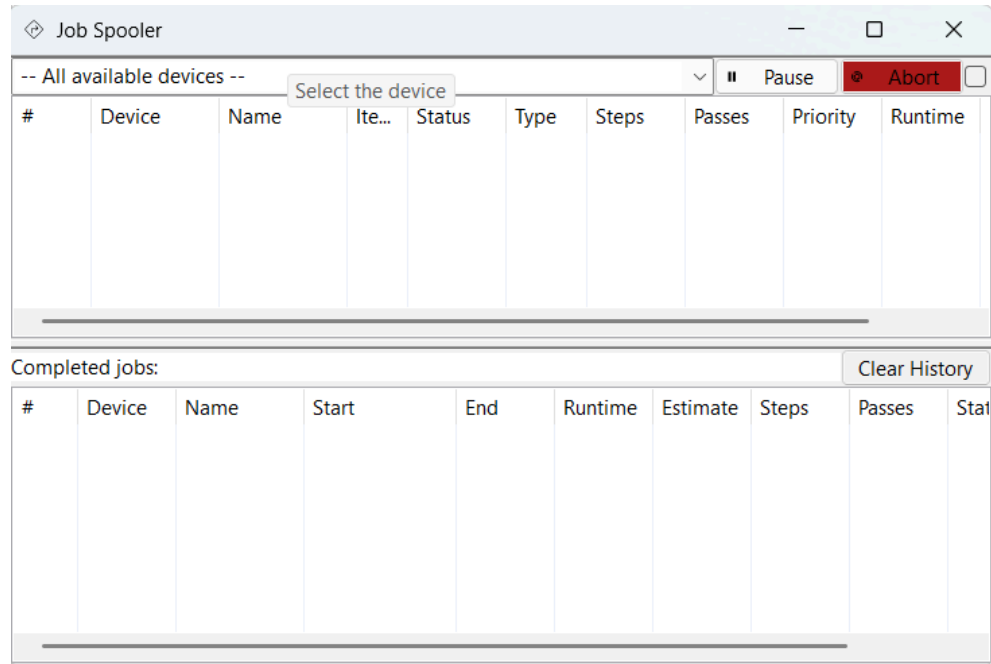
Cut plan pipeline (conceptual)

Elements + Operations

- copy → preprocess → validate → blob → [optimize] → spool
- LaserJob on device spooler

Job Spooler window

Window → **Job Spooler** — list of queued jobs, progress, cancel.



Job spooler

Simulation window

Window → **Simulation** — preview path order, travels, timing estimate.

Simulation (151.0mmx141.6mm)

Makerbase MKS DLC32

Laser Distance		Travel Distance		Total Distance	
18.69m	18.69m	7.22m (5.70m)	7.45m	25.91m	26.14m

Laser Time		Travel Time		Total Time		Extra Time	
1:49:19	1:49:19	0:01:12	0:01:14	1:50:59	1:51:01	0:00:27	0:00:27

Playback Speed: 100%

Mode: ☒ Steps ☐ Time (sec.) ☐ Time (min)

Send to Laser

Simulation

Execute Job / Thread Info

Long plans may open **Thread Info** when threaded planning is enabled — shows background planner progress.

Execute Job / Thread Info — opens during long plan builds (Tools → Burning).

14. Material Test and Material Manager

Window	Purpose
Material Manager	Library of power/speed presets per material
Material Test	Grid of test burns (power/speed matrix)

Used heavily in [17-meerkat-dlc32-workflow.md](#).

Material Manager — Tools → Editing (not captured).

Parameter-Test

Operation to test

Operation:

Cut

Show Labels / Values

☒ Labels

☒ Values

☒ Simple

First parameter (X-Axis)

Subtitle:

Parameter:

Speed

Count:

4

Every 0.667mm/s

Minimum:

1

mm/s

Maximum:

3

mm/s

Width:

10

mm

Delta:

3

mm

Color:

Red

☐ Growing

Second parameter (Y-Axis)

Subtitle:

Parameter:

Power

Count:

2

Every 400%

Minimum:

300

%

Maximum:

700

%

Height:

10

mm

Delta:

5

mm

Color:

Blue

☐ Growing

Create Pattern

Queue to Spooler

How to use it

To provide the best burning results, the parameters of operations need to be adjusted according to *YOUR* laser and the specific material you want to work with (e.g. one batch of poplar plywood from one supplier may respond completely different to a batch of another supplier despite having the very same specifications on paper).

E.g. for a regular CO2 laser you want to optimize the burn speed for a given power to reduce burn marks or decrease execution time.

Generator

Properties

Advanced

Templates

Parameter-Test

After a test grid: pick best cell → apply settings to operation → queue real job.

15. Device Manager and Configuration

Device Manager

Edit → Device-Manager — create profiles: **GRBL**, Lihuiyu, Ruida, Balor, ...

Device Manager — Edit → Device-Manager (not captured).

For DLC32 you typically have one **GRBL** device:

- Connection: **USB COM** or **TCP 192.168.x.x:8080** (WiFi bridge)
- Baud: often **115200** USB
- Options: **has endstops**, **sequential homing**, connect delay

Device Configuration

Edit → Device-Configuration — speeds, bed size, flip flags, GRBL \$ passthrough, controller behavior.

GRBL-Configuration

Device Interface Protocol Advanced Effects Operation Defaults ESP3D Upload Warning Default Actions Display

Label
GRBL-DLC32-400

Dimensions

Width 405.0mm Height 300.0mm Laserspot 0.3mm

Scale

X-Axis 1.0 Y-Axis 1.0

Flip Axis

☐ Flip X ☒ Flip Y

Axis corrections

☐ Swap XY

Home position

Force Declared Home
top-left

Z-Axis support

☐ Supports Z-axis Z-Homing \$HZ

Screen updates

☒ Device Position

Auto-Configuration

Query properties
Hardware properties

GRBL configuration — Device tab

See also §0.7 for **Advanced** and **Default Actions** screenshots.

Writing \$\$: Console shows ok per line. A **PermissionError** on “WriteFile” is MeerK40t failing to save a log on **PC** — not always a board reject. Confirm with \$\$ readback.

16. Console

Bottom **Console** pane — full command language.

Uses

- Run planning: plan0 clear copy preprocess validate blob spool
- Element ops: element merge, element subpath
- GRBL: \$101=172, \$HY, \$HX, ?, \$X unlock
- Window: window open Simulation

Console pane — open from Ribbon **Console** or Panes menu (not captured; use for \$100, \$101, \$\$).

See [14-console-quickref.md](#). Type help and help <command> for discovery.

17. Auxiliary windows (reference)

Registered in gui/wxmeerk40t.py (open via **Window** menu or console):

Window	Typical use
MeerK40t	Main frame (this manual)
Properties	Edit selection
Console	Commands
Preferences	Global settings
About	Version
Keymap	Shortcuts
Wordlist	Parametric text
MatManager	Materials library
Navigation	Jog / pulse
Notes	Project notes
AutoExec	Commands on events
JobSpooler	Queue
ThreadInfo	Planner threads
Simulation	Path preview
Tips	Daily tips

Window	Typical use
ExecuteJob	Job execution UI
BufferView	Driver buffer debug
Scene	Secondary scene view
DeviceManager	Devices
Alignment	Align objects to each other / bed
HersheyFontManager	Fonts
HersheyFontSelector	Pick font
SplitImage	Large raster tiling
OperationInfo	Operation summary
Lasertool	Laser-specific utilities
Templatetool	Templates
Hingetool	Living hinge patterns
Kerftest	Kerf compensation tests
SimpleUI	Minimal UI
CameraInterface	Camera overlay (if plugin loaded)

Alignment window — Tools → Editing (not captured).

18. View options and display modes

View → Scene Appearance toggles bits in `draw_mode`:

- Hide categories (paths, text, images)
- Hide effects / non-burn layers
- Show magnets, grid, origin
- Variable preview for wordlists

Render-Options affect how fills and vectors are interpreted at **render and burn** — only change when you understand the planner impact.

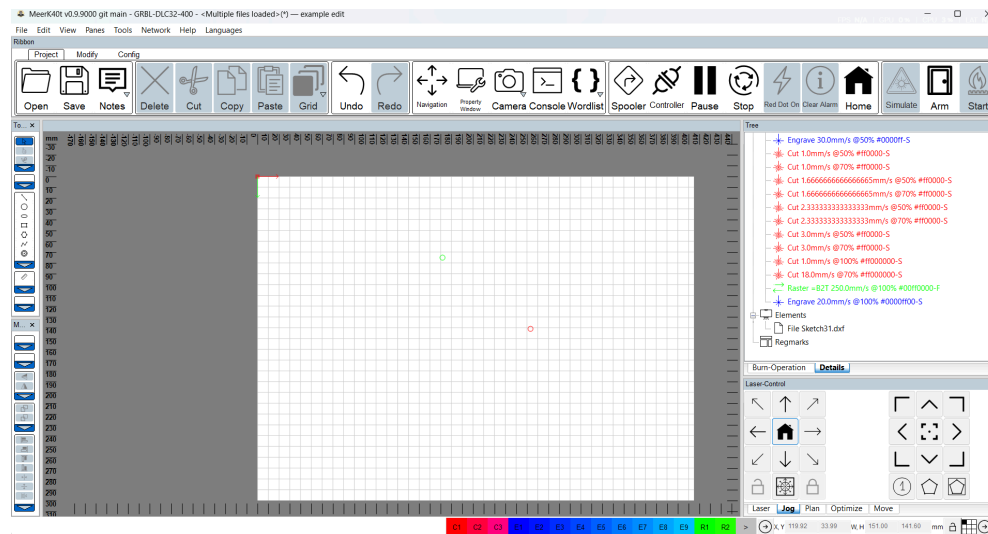
19. Network menu

Start/stop embedded servers (see §4.6). Useful for phone/tablet control on same LAN — secure your network; anyone on the port can send commands.

20. Status bar

Widgets along the bottom (gui/statusbarwidgets/):

Widget area	Info
Selection	Count, size of selection
Position	Mouse / tool position
Operation assign	Quick assign emphasized → op
Stroke/fill	Quick style readout
Job progress	Spooler step fraction during burn



Status bar and colour palette — bottom of main window

21. End-to-end workflow: import to burn

1. **File** → **Import** artwork (SVG/DXF).
 2. Check scale and position on scene.
 3. Ensure **Cut** (and **Engrave** if needed) operations exist.
 4. **Classify** or manual drag to operations.
 5. **Device-Configuration** — connect type, bed size.
 6. **Connect** on Laser pane.
 7. **Home** machine.
 8. **Pulse** / low power + jog → align to material corner.
 9. **Queue** plan (fix unassigned if warned).
 10. **Simulation** — optional but recommended for order/travel.
 11. **Arm** → **Start**.
 12. Monitor spooler; **Stop** if something wrong.
-

22. GRBL / MKS DLC32 UI notes

Specific to your setup ([16-mks-dlc32-board.md](#), [17-meerkat-dlc32-workflow.md](#)):

Topic	UI behavior
Homing	Prefer <i>HY</i> * <i>*then</i> * <i>HX</i> ; avoid single <i>SH</i>
Touch panel jog	Same GRBL \$100/\$101 as MeerK40t — calibrate with ruler
Laser test on panel	Short beam normal with \$32=1 (no motion) — use MeerK40t Pulse for steady test
WiFi	TCP device in MeerK40t; dropouts freeze progress — USB for long jobs
ALARM:1	Console \$X, check ? → Pn.; fix limits/jumpers

23. Troubleshooting the UI

Symptom	Things to check
Panes missing	View/Pane menu; Window → reset positions
Cannot click Start	Connect? Arm? Spool empty? Unassigned shapes?
Scene empty	Hide Shapes on; zoom extents; wrong emphasis
GRBL \$ won't save	One client per port; use console ok; ignore PC WriteFile error if \$\$ shows new values
Inner shapes skip	Operations tree — assign + Fill on Cut
Merge can't undo	Break Subpaths or node Break
Slow UI	Large images; suppress non-visible elements in Preferences

24. Screenshot file index

Source (your PC): C:\Users\User\Pictures\Screenshots\Meerk40T\
Copied to repo: images/meerk40t-ui-manual/

Named images (used in §0)

File	Description
01-main-window.png	Full UI, GRBL-DLC32-400, Sketch31 project
02-file-menu.png	File menu
03-ribbon-project-tab.png	Ribbon Project tab
04-scene-context-menu.png	Scene selection + right-click
05-operations-tree.png	Operations / Elements tree
07-navigation.png	Navigation window
11-grbl-config-device.png	GRBL Configuration — Device
11b-grbl-config-advanced.png	GRBL Configuration — Advanced

File	Description
11c-grbl-config-effects.png	GRBL Configuration — Effects
11d-grbl-config-default-actions.png	Default Actions at job end
11e-grbl-config-display-options.png	Display Options
13-job-spooler.png	Job Spooler
14-simulation.png	Simulation preview
16-parameter-test.png	Parameter-Test generator
18-view-menu.png	View menu
18b-gui-appearance.png	GUI Appearance submenu
18c-render-options.png	Render-Options submenu
19-edit-menu.png	Edit menu
20-tools-menu.png	Tools menu
assign-operation-submenu.png	Assign Operation
file-import-dxf.png	Import DXF dialog
kerf-test.png	Kerf-Test window
operations-insert-special.png	Operations → special commands
panes-laser-submenu.png	Panes → Laser
panes-menu-overview.png	Panes menu
panes-navigation-submenu.png	Panes → Navigation
tools-burning-submenu.png	Tools → Burning
tools-device-control.png	Tools → Device-Control

Archives (all originals)

- full-screens/ — 19 files from **Full screens**
- menu/ — 52 files from **Menu**

Still useful to capture later

Console pane, Device Manager, Material Manager, Network menu, Properties window, unassigned warning dialog, Node Edit close-up.

PDF build

```
powershell -File "C:\Users\User\Ai  
Projects\meerkat\docs\meerk40t\scripts\build-ui-manual-pdf.ps1"
```

Requires Pandoc.

Document history

Date	Change
2026-06-02	v1.0 — Initial manual, placeholders, PDF scripts
2026-06-03	v1.1 — Merged Andre screenshots (71 files); §0 illustrated walkthrough

*MeerK40t is open source; UI labels may differ slightly by version and language pack.
When in doubt, trust the on-screen text and the console help output.*